

Today's Agenda

- 1) RNN
- 2) LSTM
- 3) GRU

DL - NLP

Transformers

Computer Vision
↓
CNN

Unstructured Data
(Images / Videos)

NLP

→

Structured & Unstructured

1) Text → NER, POS, Sentiment

2) Audio

↳ Speech Recognition

Speaker Diarization

Audio Classification

3) Time Series

↳ Stock Price Prediction

4) Generation
↳ Text Generation
↳ Music Generation

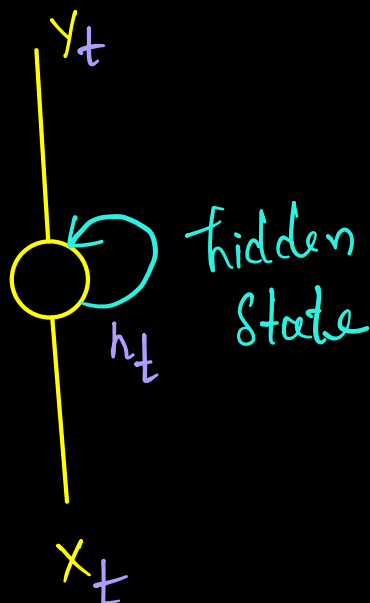
Chatbots

Sequence OR Model Sequencing

My name is Paul.

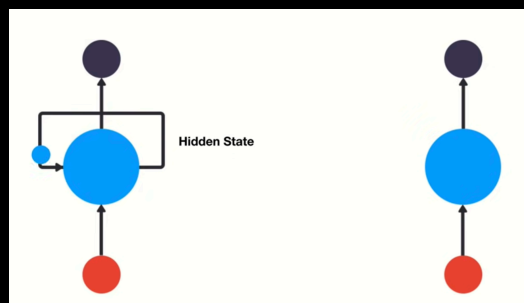
ORDER

Memory

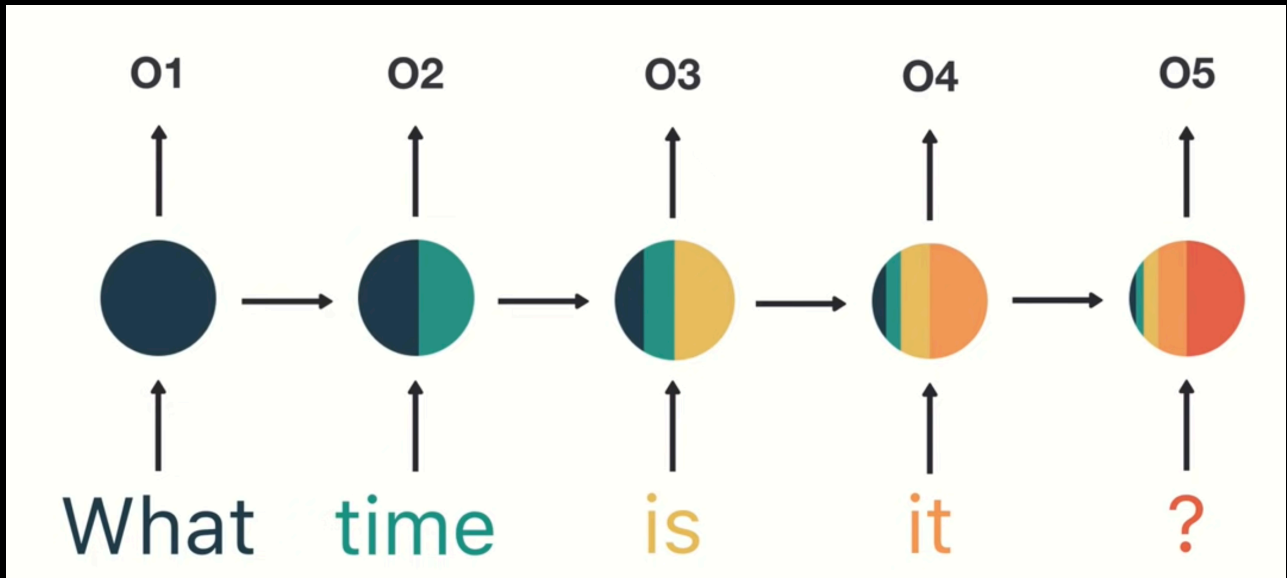


$t-1, t, t+1, t+2 \dots t_n$

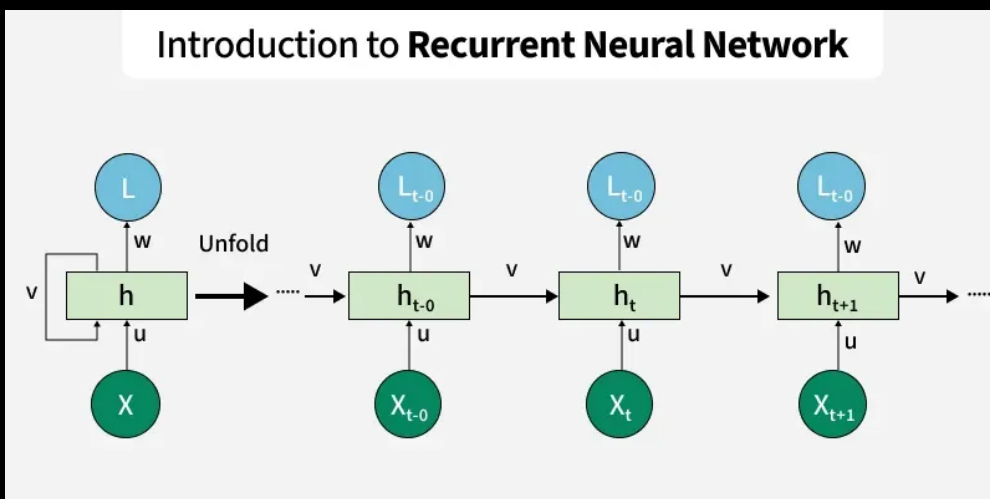
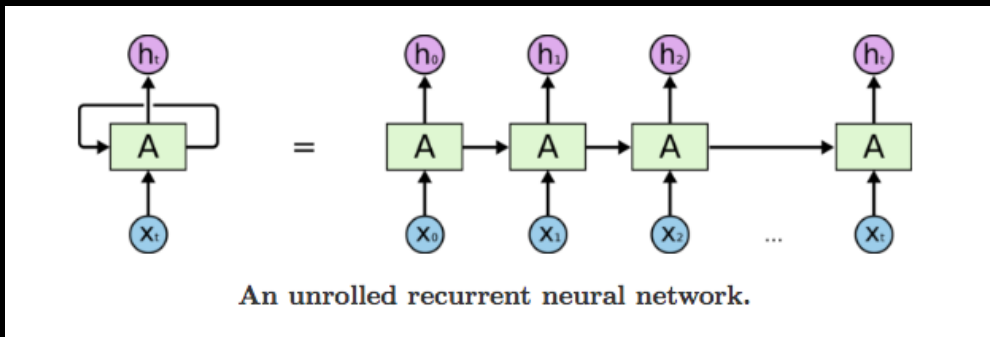
Rolled Version

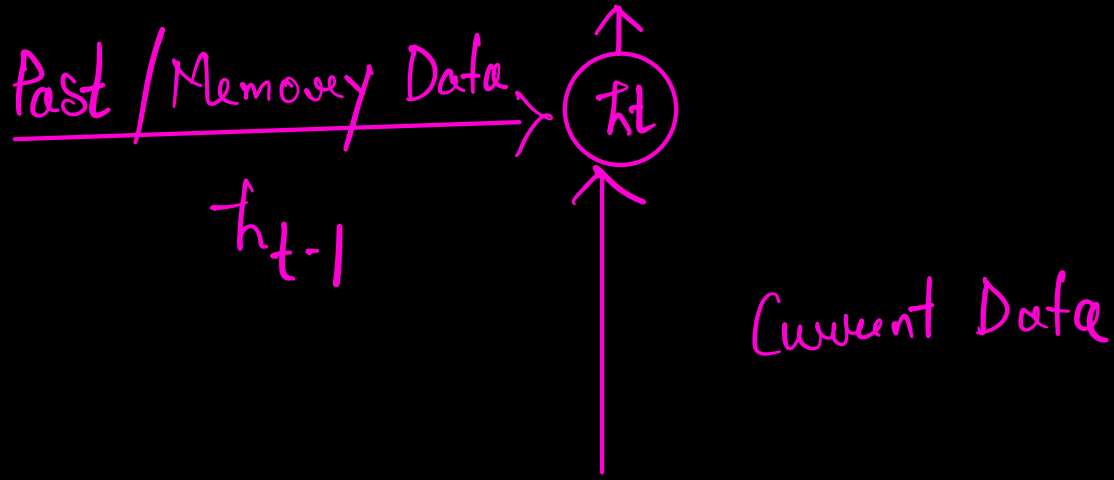


Unrolled / Unfolded version



Sequential Memory





Sequence Modelling

Types of RNN

1) One to One
One Input = One Output

Stocks
Today's $\xrightarrow{\text{Predict}}$ Tomorrow

Images
Image $\rightarrow$ Label

2) One to Many
one input $\Downarrow$ sequence
 $I \rightarrow$ Caption
Seed Word $\rightarrow$ Sentence Generation

3) Many to One
Sentiment Analysis

Fraud Analysis
Sequence $\rightarrow$ one output

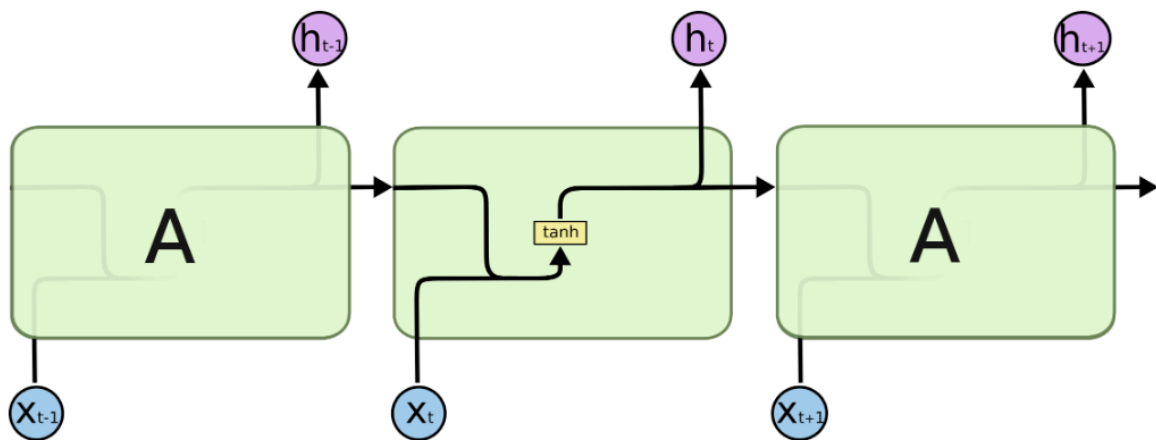
4) Many to Many

Language Translation
speech to text

RNN Cons

- 1) VG
- 2) EG
- 3) Slow Training
- 4) Very Poor Memory
- 5) Very hard to Scale

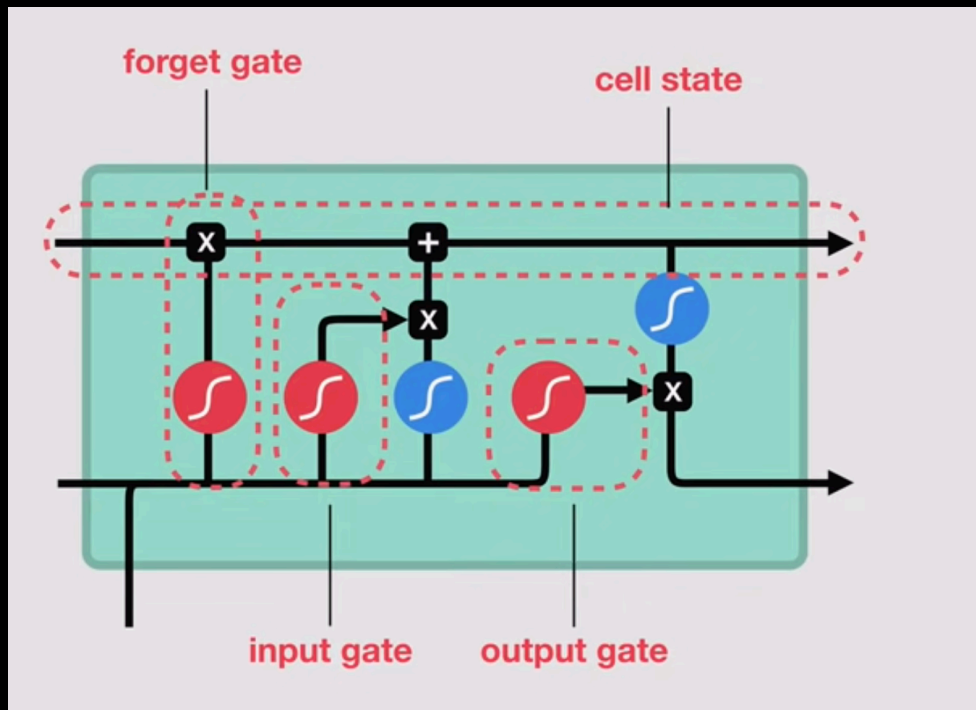
RNN $\longrightarrow$ LSTMs



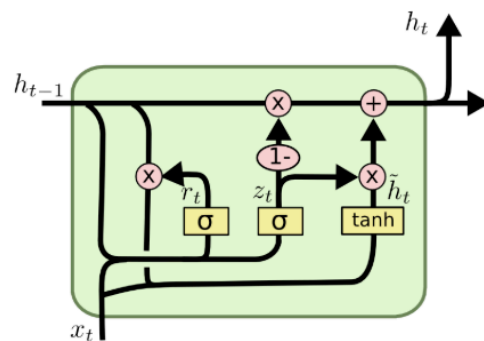
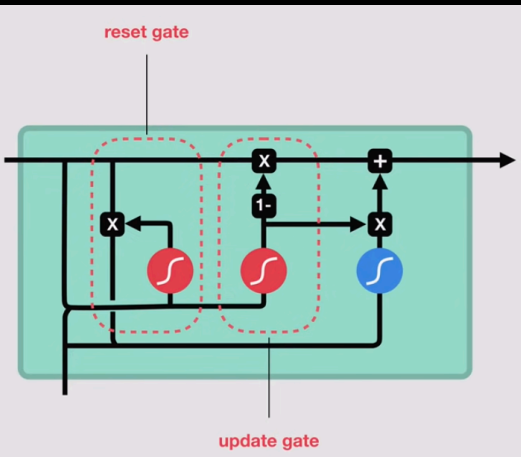
The repeating module in a standard RNN contains a single layer.

Range of $\tanh = [-1, 1]$

LSTM



GRU -> Gated Recurrent Unit



$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$$

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t])$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t])$$
$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$